Calcurion Physics Worksheets

A-Level Physics

Topic: Quantum Physics

MARKSCHEME

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1. $hf = \phi + KE_{\max}$ **A2** [2]
2. Minimum energy to free an electron from the surface **A1** [1]
3. $hf = 3.31$ eV; $KE_{\max} = 1.31$ eV **M1A1** [4]
4. Minimum frequency for photoelectric emission **A1** [1]
5. $f_0 = 4.35 \times 10^{14}$ Hz **M1A1** [3]
6. $\lambda = h/p$ **A1** [1]
7. $\lambda = 7.3 \times 10^{-10}$ m **M1A1** [3]
8. Allowed energy state of an electron in an atom **A1** [1]
9. 12.1 eV **M1A1** [2]
10. $\lambda = 103$ nm **M1A1** [3]
11. Lowest energy level ($n = 1$) **A1** [1]
12. Energy to remove electron from ground state to infinity **A1** [1]
13. Only certain ΔE allowed; $hf = \Delta E$ **A3** [3]
14. Matter and radiation exhibit both wave and particle properties **A2** [2]
15. Electron diffraction / Davisson–Germer **A1** [1]
16. Threshold frequency; instantaneous emission; KE depends on f not intensity **A3** [3]
17. $hf = 4.97$ eV; $KE_{\max} = 1.97$ eV; $V_s = 1.97$ V **M1A1** [4]
18. $p = \sqrt{2mE}$; $\lambda = 1.74 \times 10^{-10}$ m **M1A1** [4]
19. Levels; Lyman to $n = 1$; Balmer to $n = 2$ **A3** [3]
20. $\Delta E = 2.55$ eV; $f = 6.16 \times 10^{14}$ Hz **M1A1** [3]
21. Emission: bright lines; absorption: dark lines at same wavelengths **A3** [3]
22. $p = E/c = 1.33 \times 10^{-27}$ kg m/s **M1A1** [3]
23. Intensity increases number of photons, not energy per photon **A2** [2]
24. Electrons diffract through crystal; wave nature **A3** [3]
25. $\lambda_{\min} = hc/eV = 4.97 \times 10^{-11}$ m **M1A1** [3]